

네트워크 실무

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1. SSLVPN 개요

◆ SSL VPN 개요

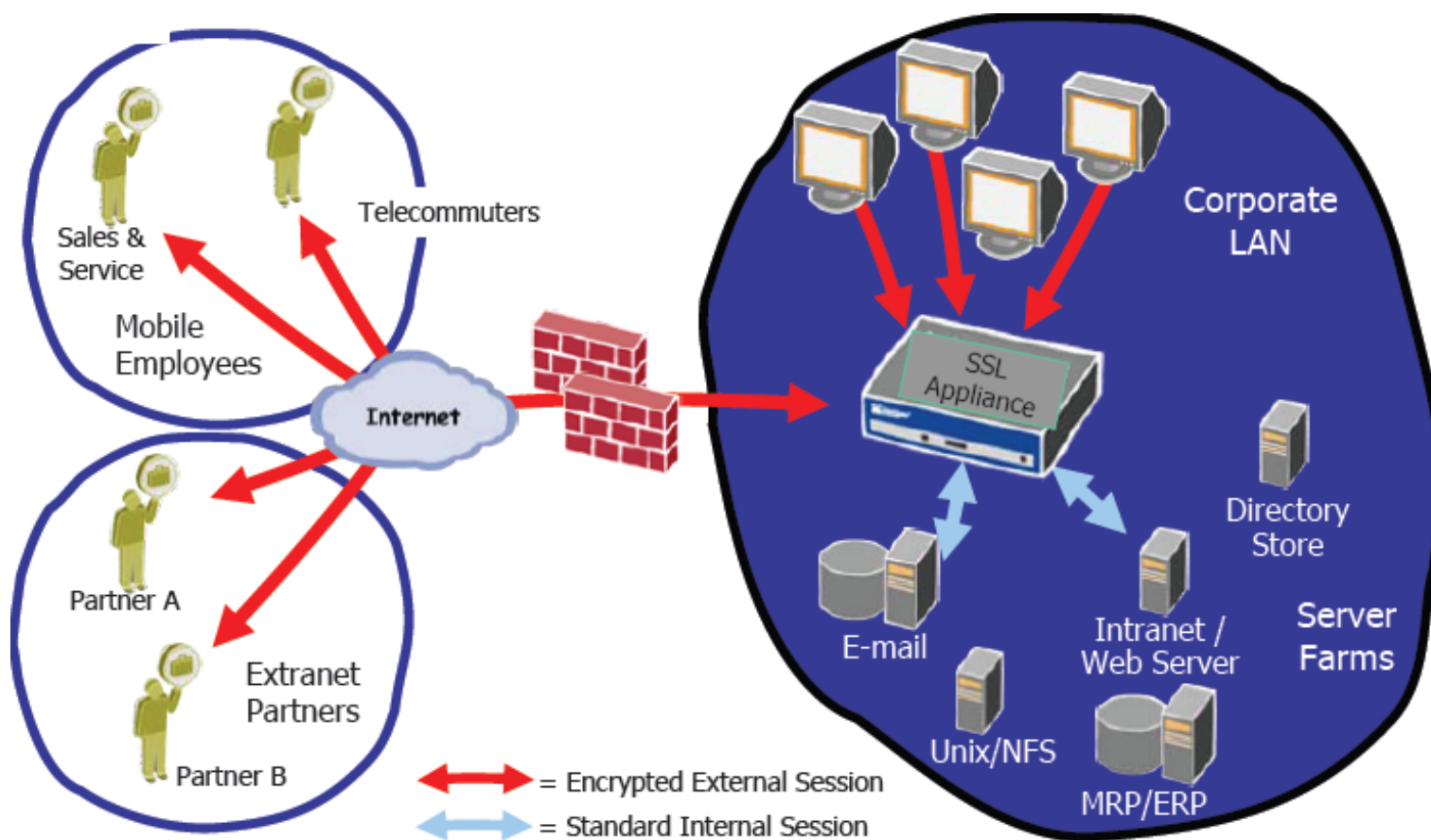
- SSL(SecureSockets Layer)을 이용한 가상사설망(VPN)으로서, 인터넷 익스플로러, 넷스케이프 내비게이터와 같은 기본 웹브라우저에서 이용 가능한 SSL을 활용해 손쉽게 보안성이 높은 VPN을 구성 하는 솔루션임.
- 기존 VPN 구성에서는 원격 사용자의 데스크톱 PC나 노트북 PC에 반드시 클라이언트 소프트웨어가 깔려 있어야 VPN에 접속할 수 있지만, SSL VPN은 본사에 SSL 전용장비를 설치하고 클라이언트에는 기본 웹브라우저만 있으면 어디서든 간편하게 접속할 수 있는것이 강점임
- 기존 VPN이 갖고 있던 클라이언트 소프트웨어의 버전과 패치관리의 어려움, 사용자의 다양한 하드웨어, 운용체계 버전 및 패치상태에 따른 설치 어려움 등을 효과적으로 해결할 수 있음

◆ SSL VPN 장점

- Web 과 non-web 환경 모두에 하드웨어 베이스 암호화 지원(VPN)
- 파일 레벨까지 제어할 수 있는 유연한 access control
- Application, group, role별 사용자 접근 제어(EAM)
- 인증서를 이용한 강력한 사용자 인증 방식
- Application 서버의 proxy역할로 웹서버의 해킹위험 감소
- 인터넷을 통해 내부 자원에 접근하려는 모든 요청 처리
- 서버의 H/W나 S/W 수정 없이 TLS/SSL에 대한 보안 Web Server를 구현

SSL VPN

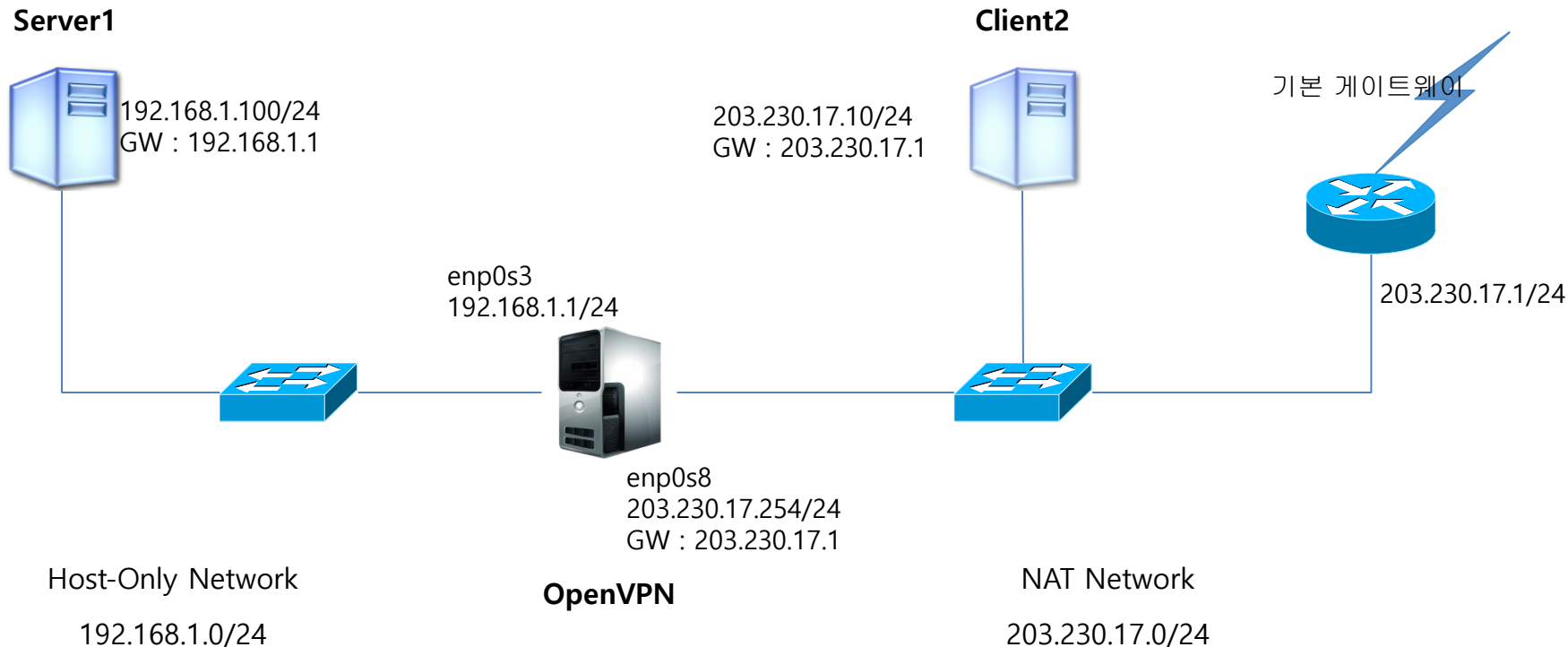
◆ SSL VPN 구성 및 활용방법



2. OPENVPN 설치

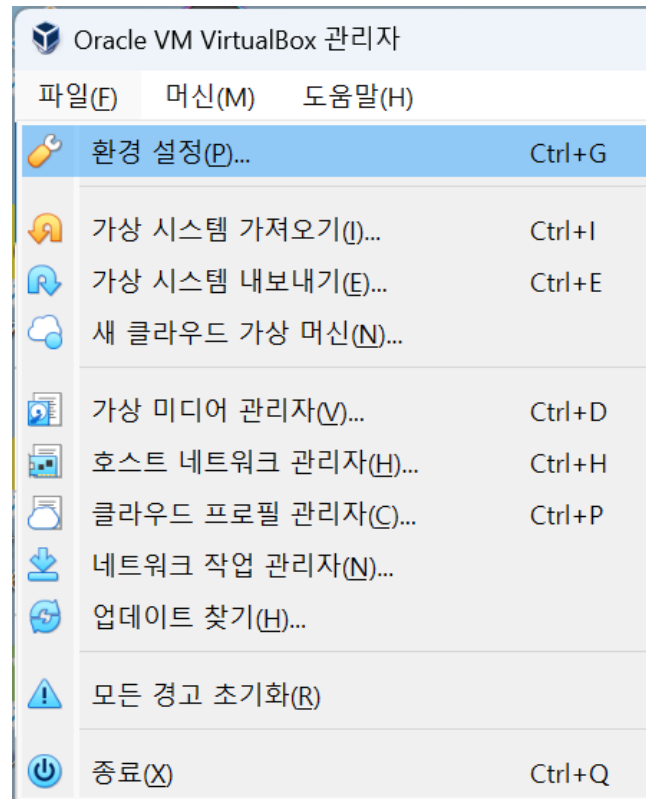
구성도

- 우분투 리눅스로 방화벽을 만들어 서로 다른 네트워크간 ACL 통신을 할 수 있도록 설정하도록 합니다.



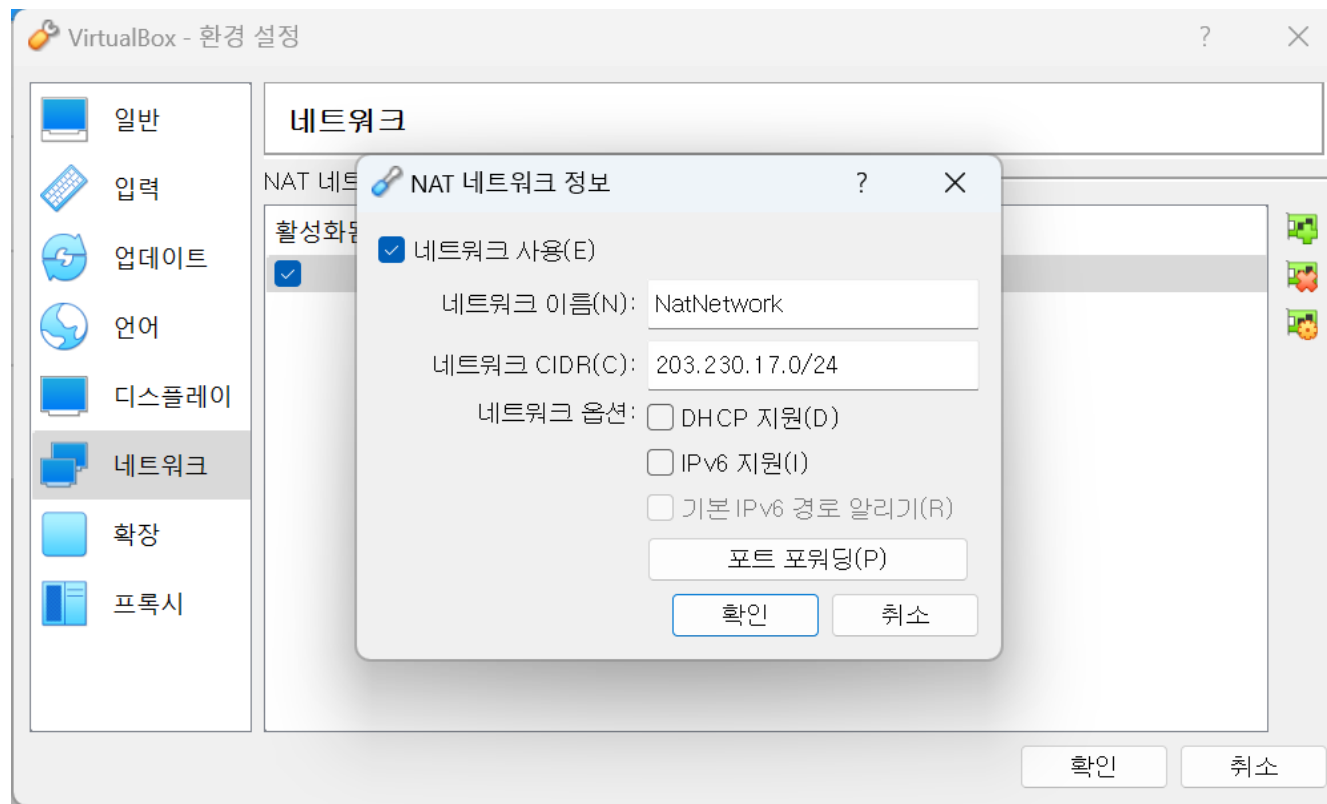
Network 환경 설정

- VirtualBox 프로그램에서 '파일(F)' → '환경설정(P)...'를 실행한다.



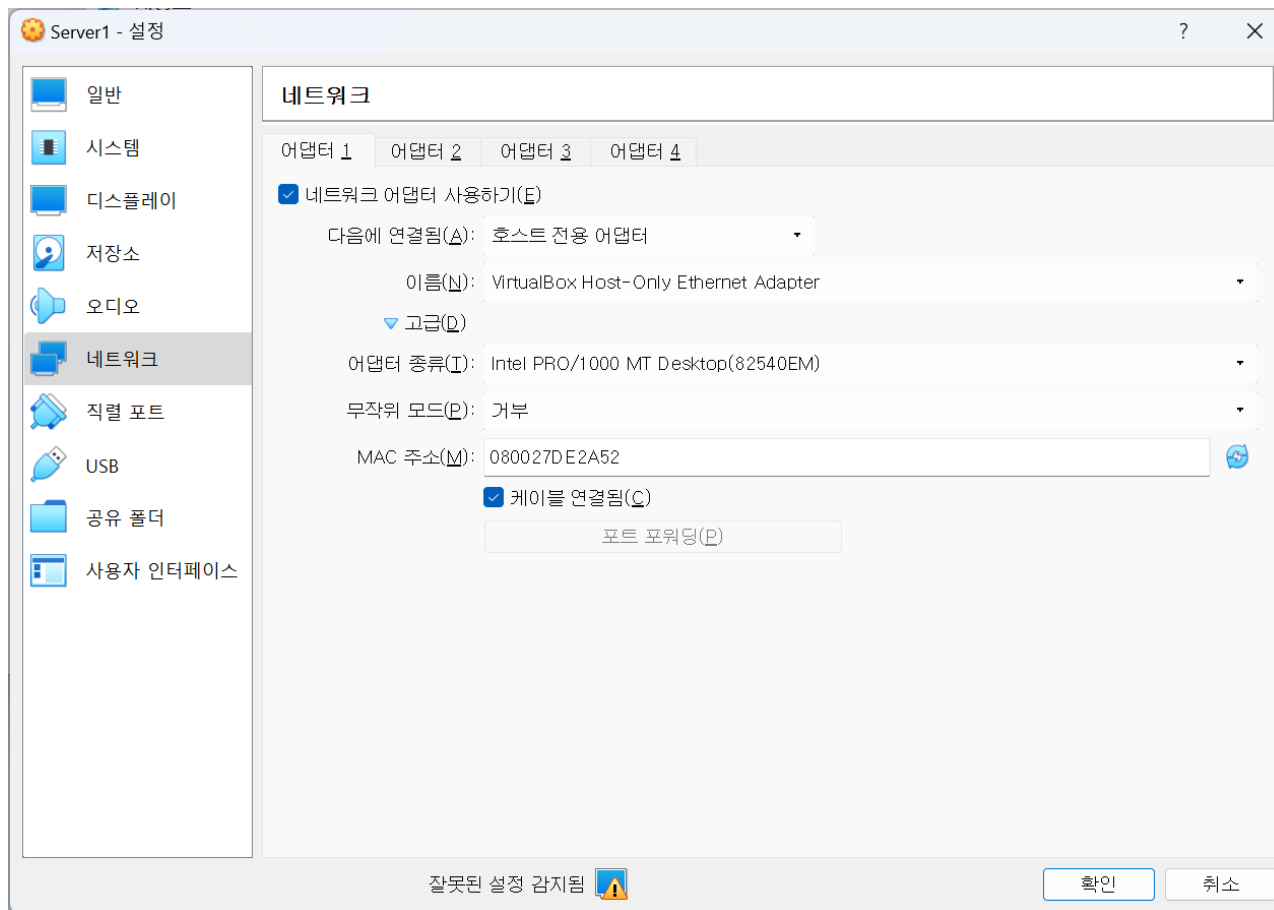
Network 환경 설정

- 네트워크를 선택하고 NAT 네트워크를 추가한다.
 - 네트워크 CIDR© : 203.230.17.0/24
 - DHCP지원 → 제거



Server1 설정

■ Network Adapter : VirtualBox Host-Only Ethernet Adapter



Server1 설정

■ Ubuntu Network 설정

- IPv4 Method : Manual
- IP : 192.168.1.10
- Netmask : 24
- Gateway : 192.168.1.1
- DNS : 8.8.8.8

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
192.168.1.10	255.255.255.0	192.168.1.1	

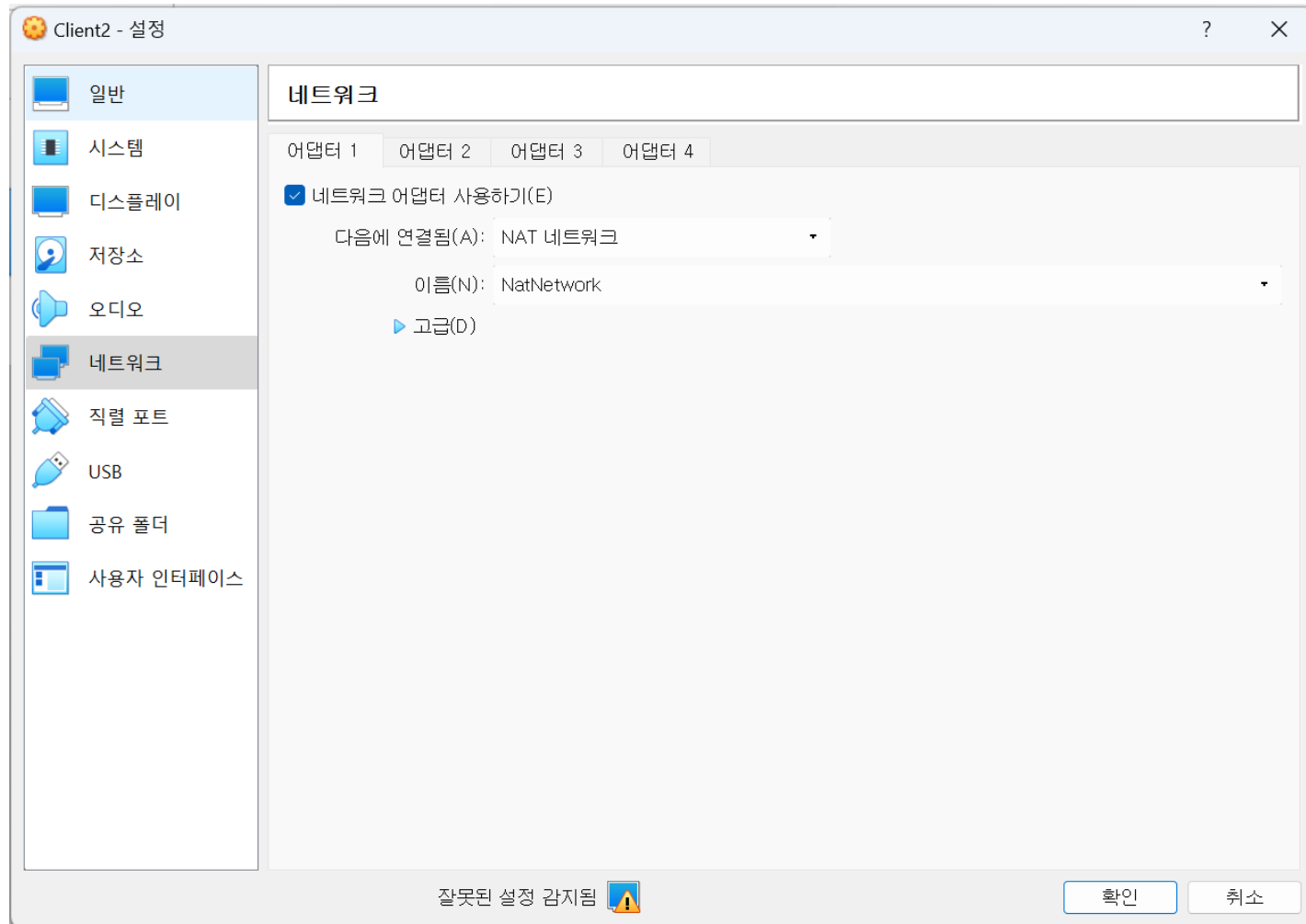
DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

Client2 설정

- Network Adapter : NAT 네트워크 -> NATNetwork 선택



Client2 설정

■ Ubuntu Network 설정

- IPv4 Method : Manual
- IP : 203.230.17.10
- Netmask : 24
- Gateway : 203.230.17.1
- DNS : 8.8.8.8

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
203.230.17.10	255.255.255.0	203.230.17.1	

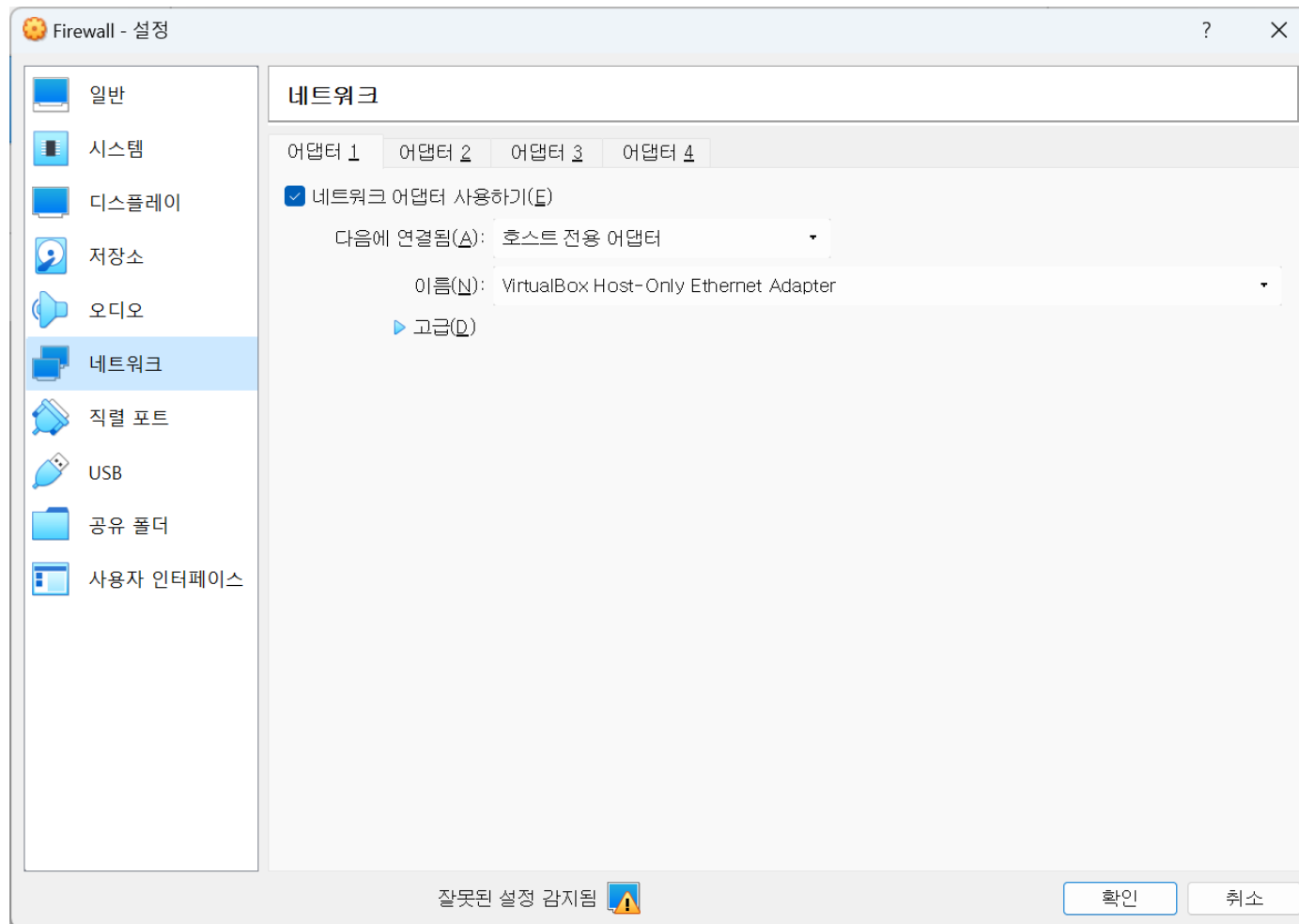
DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

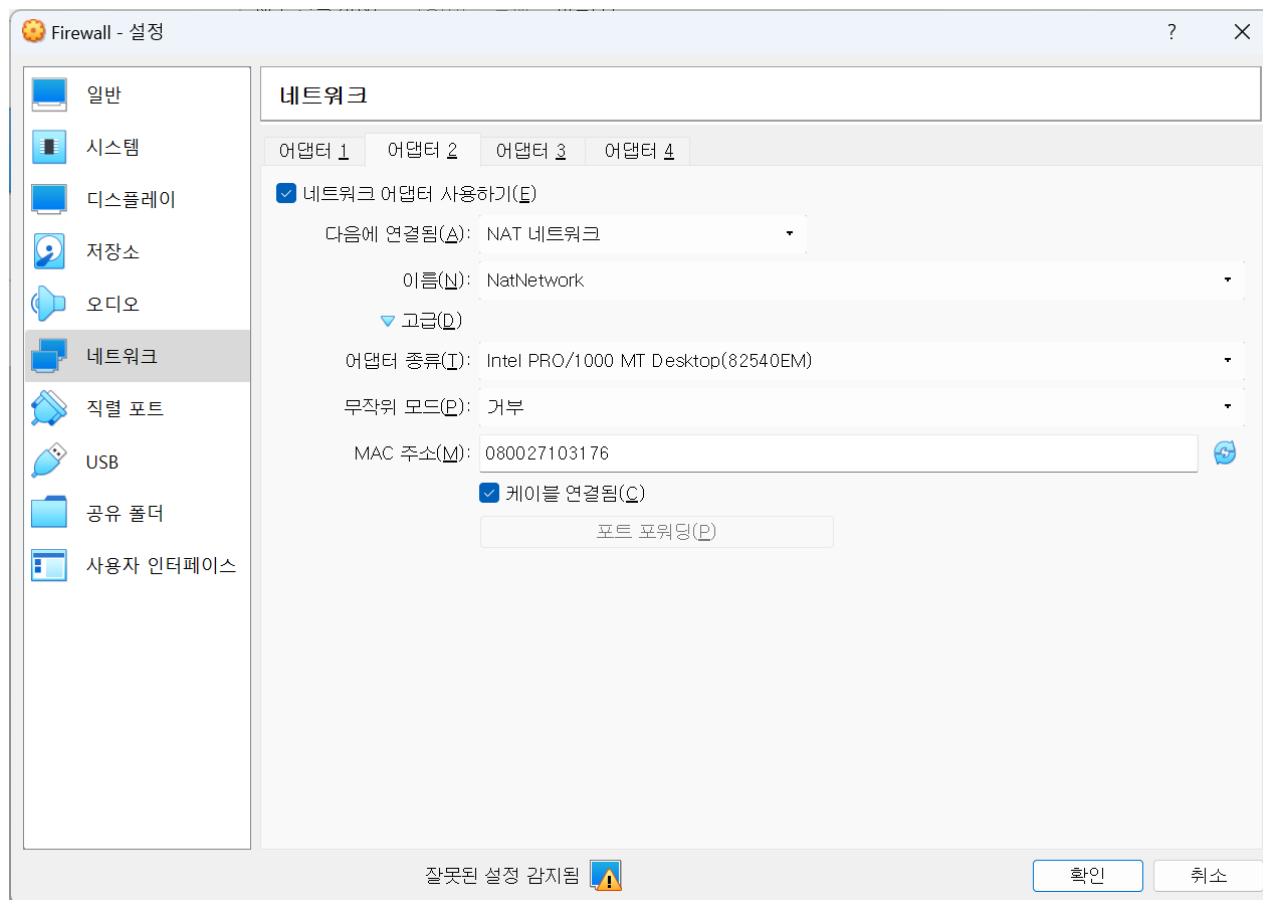
OpenVPN 설정

- Network Adapter : VirtualBox Host-Only Ethernet Adapter



OpenVPN 설정

- Network Adapter 2 : NAT 네트워크 -> NATNetwork 선택



OpenVPN 설정

■ Ubuntu Network 설정

- Interface : enp0s3
- IPv4 Method : Manual
- IP : 192.168.1.1
- Netmask : 24
- DNS : 8.8.8.8

The screenshot shows the 'Wired' network settings window in Ubuntu. The 'IPv4' tab is selected, showing the following configuration:

- IPv4 Method:** Manual (selected), Automatic (DHCP), Link-Local Only, Disable, Shared to other computers.
- Addresses:** A table with columns for Address, Netmask, and Gateway. The first row shows 192.168.1.1 and 255.255.255.0.
- DNS:** 8.8.8.8 (with an 'Automatic' toggle switch).

Address	Netmask	Gateway
192.168.1.1	255.255.255.0	

OpenVPN 설정

■ Ubuntu Network 2 설정

- Interface : enp0s8
- IPv4 Method : Manual
- IP : 203.230.17.254
- Netmask : 24
- Gateway : 203.230.17.1
- DNS : 8.8.8.8

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ **Manual** ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
203.230.17.254	255.255.255.0	203.230.17.1	

DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

OpenVPN 설정

■ Interface간 패킷 전달

➤ enp0s3 Interface와 enp0s85 Interface간 패킷 전송을 위한 Routing 기능 활성화

➤ \$ sudo vi /etc/sysctl.conf

```
# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=0

# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
```

➤ #net.ipv4.ip_forwarding=1 행을 찾아서 맨 앞의 # 제거 후 저장

➤ 변경된 내용을 적용

➤ \$ sudo sysctl -p /etc/sysctl.conf

```
ubuntu@Firewall:~/Desktop$ sudo sysctl -p /etc/sysctl.conf
[sudo] password for ubuntu:
net.ipv4.ip_forward = 1
ubuntu@Firewall:~/Desktop$
```

OpenVPN을 위한 공인인증서 패키지 설치

■ RSA 패키지 설치

➤ apt install easy-rsa -y

```
root@Firewall: ~  
root@Firewall:~# apt install easy-rsa  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  libfwupdplugin1 libxmlb1  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  libccid opencsc opencsc-pkcs11 pcscd  
The following NEW packages will be installed:  
  easy-rsa libccid opencsc opencsc-pkcs11 pcscd  
0 upgraded, 5 newly installed, 0 to remove and 33 not upgraded.  
Need to get 1,343 kB of archives.  
After this operation, 4,992 kB of additional disk space will be used.  
Do you want to continue? [Y/n]
```

인증서 생성

■ CA(Certificate Authority) 생성

```
root@OpenVPN: ~/easy-rsa

root@OpenVPN:~# make-cadir easy-rsa
root@OpenVPN:~# ls
easy-rsa  snap
root@OpenVPN:~# cd easy-rsa/
root@OpenVPN:~/easy-rsa# cp vars vars.orig
root@OpenVPN:~/easy-rsa# gedit vars&
[1] 4141
root@OpenVPN:~/easy-
```

```
root@OpenVPN: ~/easy-rsa

GNU nano 4.8                                vars                                Modified
# These are the default values for fields which will be placed in the
# certificate.  Don't leave any of these fields blank, although interactively
# you may omit any specific field by typing the "." symbol (not valid for
# email.)

set_var EASYRSA_REQ_COUNTRY    "US"
set_var EASYRSA_REQ_PROVINCE   "NY"
set_var EASYRSA_REQ_CITY       "NY"
set_var EASYRSA_REQ_ORG        "TEST"
set_var EASYRSA_REQ_EMAIL      "test@test.com"
set_var EASYRSA_REQ_OU         "TEST"

# Choose a size in bits for your keypairs. The recommended value is 2048. Using
# 2048-bit keys is considered more than sufficient for many years into the
# future. Larger key sizes will slow down TLS negotiation and make key/DH param
# generation take much longer. Values up to 4096 should be accepted by most
# software. Only used when the crypto alg is rsa (see below.)

#set_var EASYRSA_KEY_SIZE      2048
```

➤ 내용 변경

인증서 생성

■ CA(Certificate Authority) 생성

➤ CA Key 패스워드 입력 : infosec_01

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# ls -l
total 32
lrwxrwxrwx 1 root root 27 12월 11 22:18 easyrsa -> /usr/share/easy-rsa/easyrsa
-rw-r--r-- 1 root root 4651 12월 11 22:18 openssl-easyrsa.cnf
-rw-r--r-- 1 root root 8514 12월 11 22:32 vars
-rw-r--r-- 1 root root 8576 12월 11 22:19 vars.orig
lrwxrwxrwx 1 root root 30 12월 11 22:18 x509-types -> /usr/share/easy-rsa/x509-types
root@OpenVPN:~/easy-rsa# ./easyrsa init-pki

Note: using Easy-RSA configuration from: ./vars

init-pki complete; you may now create a CA or requests.
Your newly created PKI dir is: /root/easy-rsa/pki

root@OpenVPN:~/easy-rsa# ./easyrsa build-ca

Note: using Easy-RSA configuration from: ./vars

Using SSL: openssl OpenSSL 1.1.1f 31 Mar 2020

Enter New CA Key Passphrase:
Re-Enter New CA Key Passphrase:
Generating RSA private key, 2048 bit long modulus (2 primes)
```

인증서 생성

■ CA(Certificate Authority) 생성

- CA 이름 : [Enter]
- CA인증서 복사

```
root@OpenVPN: ~/easy-rsa
Enter New CA Key Passphrase:
Re-Enter New CA Key Passphrase:
Generating RSA private key, 2048 bit long modulus (2 primes)
.....+++++
.....+++++
e is 65537 (0x010001)
Can't load /root/easy-rsa/pki/.rnd into RNG
140589764666688:error:2406F079:random number generator:RAND_load_file:Cannot open file:../crypto/rand/randfile.c:98:Filename=/root/easy-rsa/pki/.rnd
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:

CA creation complete and you may now import and sign cert requests.
Your new CA certificate file for publishing is at:
/root/easy-rsa/pki/ca.crt

root@OpenVPN:~/easy-rsa# cp ./pki/ca.crt ~/
root@OpenVPN:~/easy-rsa#
```

서버용 Key 생성

■ 서버용 Key 생성

➤ 서버용 이름 test로 생성

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# ./easyrsa gen-req test nopass

Note: using Easy-RSA configuration from: ./vars

Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020
Generating a RSA private key
.....+++++
...+++++
writing new private key to '/root/easy-rsa/pki/private/test.key.mQMzkTwWyF'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [test]:

Keypair and certificate request completed. Your files are:
req: /root/easy-rsa/pki/reqs/test.req
key: /root/easy-rsa/pki/private/test.key

root@OpenVPN:~/easy-rsa#
```

서버용 Key 생성

■ 서버용 Key 생성

- Test 서버용 인증서 생성
- Confirm에 'yes' 입력
- ca.key 비밀번호 입력 :
infosec_01

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# ./easyrsa sign-req server test

Note: using Easy-RSA configuration from: ./vars
Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020

You are about to sign the following certificate.
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.

Request subject, to be signed as a server certificate for 1080 days:

subject=
  commonName                = test

Type the word 'yes' to continue, or any other input to abort.
Confirm request details: yes
Using configuration from /root/easy-rsa/pki/safessl-easyrsa.cnf
Enter pass phrase for /root/easy-rsa/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName             :ASN.1 12:'test'
Certificate is to be certified until Nov 25 13:44:22 2025 GMT (1080 days)

Write out database with 1 new entries
Data Base Updated

Certificate created at: /root/easy-rsa/pki/issued/test.crt
```


서버용 Key 생성

■ 서버용 Key 생성

➤ 생성된 서버의 키와 인증서 복사

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# cp ./pki/private/test.key ./pki/issued/test.crt ~/
root@OpenVPN:~/easy-rsa# ls -l ~/
total 24
-rw----- 1 root root 1204 12월 11 22:34 ca.crt
drwx----- 3 root root 4096 12월 11 22:33 easy-rsa
drwx----- 3 root root 4096 10월 30 17:40 snap
-rw----- 1 root root 4600 12월 11 22:47 test.crt
-rw----- 1 root root 1704 12월 11 22:47 test.key
root@OpenVPN:~/easy-rsa#
```

DH(Diffie-Hellman) 생성

- DH(Diffie-Hellman) 생성
- 생성된 DH 복사

```

root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# ./easyrsa gen-dh

Note: using Easy-RSA configuration from: ./vars

Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020
Generating DH parameters, 2048 bit long safe prime, generator 2
This is going to take a long time
.....
...+.....+.....+.....+.....
.....+.....
.....+.....+.....+.....+
*++*++*++*

DH parameters of size 2048 created at /root/easy-rsa/pki/dh.pem

root@OpenVPN:~/easy-rsa# cp ./pki/dh.pem ~/dh2048.pem
root@OpenVPN:~/easy-rsa# ls -l ~/
total 28
-rw----- 1 root root 1204 12월 11 22:34 ca.crt
-rw----- 1 root root  424 12월 11 22:51 dh2048.pem
drwx----- 3 root root 4096 12월 11 22:33 easy-rsa
drwx----- 3 root root 4096 10월 30 17:40 snap
-rw----- 1 root root 4600 12월 11 22:47 test.crt
-rw----- 1 root root 1704 12월 11 22:47 test.key
root@OpenVPN:~/easy-rsa#

```

SSL 통신용 인증키 생성

■ tls-auth key (ta.key) 생성

➤ 생성된 인증용 키 복사

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# openssl genpkey --genkey --secret ta.key
root@OpenVPN:~/easy-rsa# cp ./ta.key ~/
root@OpenVPN:~/easy-rsa# ls -l ~/
total 32
-rw----- 1 root root 1204 12월 11 22:34 ca.crt
-rw----- 1 root root 424 12월 11 22:51 dh2048.pem
drwx----- 3 root root 4096 12월 11 22:55 easy-rsa
drwx----- 3 root root 4096 10월 30 17:40 snap
-rw----- 1 root root 636 12월 11 22:55 ta.key
-rw----- 1 root root 4600 12월 11 22:47 test.crt
-rw----- 1 root root 1704 12월 11 22:47 test.key
root@OpenVPN:~/easy-rsa#
```

서버 정보 설정

■ 서버 환경 설정

- Server용 샘플 파일 복사
- 서버 이름인 test.conf로 복사
- 내용 변경 후 복사

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# cp /usr/share/doc/openvpn/examples/sample-config-files/server.conf.gz ./
root@OpenVPN:~/easy-rsa# gzip -d server.conf.gz
root@OpenVPN:~/easy-rsa# cp server.conf test.conf
root@OpenVPN:~/easy-rsa# nano test.conf
```

```
GNU nano 4.8 test.conf Modified
# Any X509 key management system can be used.
# OpenVPN can also use a PKCS #12 formatted key file
# (see "pkcs12" directive in man page).
ca ca.crt
cert test.crt
key test.key # This file should be kept secret

# Diffie hellman parameters.
# Generate your own with:
# back to the OpenVPN server.
push "route 192.168.1.0 255.255.255.0"
;push "route 192.168.20.0 255.255.255.0"
```

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# cp test.conf ~/
root@OpenVPN:~/easy-rsa# la -l
total 64
lrwxrwxrwx 1 root root 27 12월 11 22:18 easyrsa -> /usr/share/easy-rsa/easyrsa
-rw-r--r-- 1 root root 4651 12월 11 22:18 openssl-easyrsa.cnf
```

클라이언트용 Key 생성

■ 클라이언트용 Key 생성

- 클라이언트 이름으로 ubuntu01의 키 생성

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# ./easyrsa gen-req ubuntu01 nopass

Note: using Easy-RSA configuration from: ./vars

Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020
Generating a RSA private key
.....+++++
...+++++
writing new private key to '/root/easy-rsa/pki/private/ubuntu01.key.IClw1xyI7i'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [ubuntu01]:

Keypair and certificate request completed. Your files are:
req: /root/easy-rsa/pki/reqs/ubuntu01.req
key: /root/easy-rsa/pki/private/ubuntu01.key

root@OpenVPN:~/easy-rsa#
```

클라이언트용 Key 생성

■ 클라이언트용 Key 생성

- ubuntu01 클라이언트용 인증서 생성
- Confirm에 'yes' 입력
- ca.key 패스워드 입력 :
infosec_01

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# ./easyrsa sign-req client ubuntu01
Note: using Easy-RSA configuration from: ./vars
Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020

You are about to sign the following certificate.
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.

Request subject, to be signed as a client certificate for 1080 days:

subject=
  commonName                = ubuntu01

Type the word 'yes' to continue, or any other input to abort.
Confirm request details: yes
Using configuration from /root/easy-rsa/pki/safessl-easyrsa.cnf
Enter pass phrase for /root/easy-rsa/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName      :ASN.1 12:'ubuntu01'
Certificate is to be certified until Nov 25 14:12:08 2025 GMT (1080 days)

Write out database with 1 new entries
Data Base Updated

Certificate created at: /root/easy-rsa/pki/issued/ubuntu01.crt
```

클라이언트용 버용 Key 생성

■ 클라이언트용 Key 생성

➤ 생성된 클라이언트용 키와 인증서 복사

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# cp ./pki/private/ubuntu01.key ./pki/issued/ubuntu01.crt ~/
root@OpenVPN:~/easy-rsa# ls -l ~/
total 56
-rw----- 1 root root 1204 12월 11 22:34 ca.crt
-rw----- 1 root root 424 12월 11 22:51 dh2048.pem
drwx----- 3 root root 4096 12월 11 23:04 easy-rsa
drwx----- 3 root root 4096 10월 30 17:40 snap
-rw----- 1 root root 636 12월 11 22:55 ta.key
-rw-r--r-- 1 root root 10848 12월 11 23:04 test.conf
-rw----- 1 root root 4600 12월 11 22:47 test.crt
-rw----- 1 root root 1704 12월 11 22:47 test.key
-rw----- 1 root root 4498 12월 11 23:14 ubuntu01.crt
-rw----- 1 root root 1704 12월 11 23:14 ubuntu01.key
root@OpenVPN:~/easy-rsa#
```

클라이언트 정보 설정

■ 클라이언트 환경 설정

- Client용 샘플 파일 복사
- 클라이언트 이름인 ubuntu01.ovpn으로 복사 후 내용 변경

```
root@OpenVPN: ~/easy-rsa
root@OpenVPN:~/easy-rsa# cp /usr/share/doc/openvpn/examples/sample-config-files/client.conf ./
root@OpenVPN:~/easy-rsa# cp client.conf ~/ubuntu01.ovpn
root@OpenVPN:~/easy-rsa# nano ~/ubuntu01.ovpn
```

```
root@OpenVPN: ~/easy-rsa
GNU nano 4.8 /root/ubuntu01.ovpn
# The hostname/IP and port of the server.
# You can have multiple remote entries
# to load balance between the servers.
remote 203.230.17.254 1194
;remote my-server-2 1194

# a separate .crt/.key file pair
# for each client. A single ca
# file can be used for all clients.
ca ca.crt
cert ubuntu01.crt
key ubuntu01.key
```


OpenVPN 서버 서비스 기동

- OpenVPN 실행에 필요한 키 및 인증서 복사
- OpenVPN을 test란 이름의 서버로 실행

```
root@OpenVPN: ~  
root@OpenVPN:~/easy-rsa# cd ~/  
root@OpenVPN:~# ls  
ca.crt      easy-rsa  ta.key      test.crt   ubuntu01.crt  ubuntu01.ovpn  
dh2048.pem  snap      test.conf   test.key   ubuntu01.key  
root@OpenVPN:~# ls /etc/openvpn/  
client  server  update-resolv-conf  
root@OpenVPN:~# cp ca.crt dh2048.pem ta.key test*.*/etc/openvpn  
root@OpenVPN:~# ls /etc/openvpn/  
ca.crt  client  dh2048.pem  server  ta.key  test.conf  test.crt  test.key  update-resolv-conf  
root@OpenVPN:~# systemctl start openvpn@test  
root@OpenVPN:~# journalctl -u openvpn@test -xe  
-- Defined-By: systemd  
-- Support: http://www.ubuntu.com/support  
--  
-- A start job for unit openvpn@test.service has begun execution.  
--  
-- The job identifier is 3845.  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: OpenVPN 2.4.7 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [V  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: library versions: OpenSSL 1.1.1f 31 Mar 2020, LZO 2.10  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: Diffie-Hellman initialized with 2048 bit key  
12월 11 23:35:31 OpenVPN systemd[1]: Started OpenVPN connection to test.  
-- Subject: A start job for unit openvpn@test.service has finished successfully  
-- Defined-By: systemd  
-- Support: http://www.ubuntu.com/support  
--  
-- A start job for unit openvpn@test.service has finished successfully.  
--  
-- The job identifier is 3845.  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: Outgoing Control Channel Authentication: Using 160 bit me  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: Incoming Control Channel Authentication: Using 160 bit me  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: ROUTE_GATEWAY 203.230.17.1/255.255.255.0 IFACE=enp0s8 HWA  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: TUN/TAP device tun0 opened  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: TUN/TAP TX queue length set to 100  
12월 11 23:35:31 OpenVPN ovpn-test[5282]: /sbin/ip link set dev tun0 up mtu 1500
```

OpenVPN 통신

■ 통신 테스트

➤ Client2 → Server1 로 통신 요청

```
ubuntu@client2: ~  
ubuntu@client2:~$ ping 192.168.1.10 -c 2  
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.  
  
--- 192.168.1.10 ping statistics ---  
2 packets transmitted, 0 received, 100% packet loss, time 1103ms  
  
ubuntu@client2:~$
```

➤ Client1 → Server2 로 통신 요청

```
ubuntu@Server1: ~  
ubuntu@Server1:~$ ping 203.230.17.10 -c 2  
PING 203.230.17.10 (203.230.17.10) 56(84) bytes of data.  
64 bytes from 203.230.17.10: icmp_seq=1 ttl=63 time=0.545 ms  
64 bytes from 203.230.17.10: icmp_seq=2 ttl=63 time=0.471 ms  
  
--- 203.230.17.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1027ms  
rtt min=0.471 ms max=0.545 ms avg=0.508 ms
```

```
root@client2: /home/ubuntu  
root@client2:/home/ubuntu# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
23:44:50.754363 IP 203.230.17.254 > client2: ICMP echo request, id 2, seq 1, length 64  
23:44:50.754382 IP client2 > 203.230.17.254: ICMP echo reply, id 2, seq 1, length 64
```

OpenVPN 통신

■ 클라이언트 OpenVPN 설정

- OpenVPN 서버에 접속하여 클라이언트 실행에 필요한 파일들 복사

```
ubuntu@client2: ~  
ubuntu@client2:~$ sftp ubuntu@203.230.17.254  
ubuntu@203.230.17.254's password:  
Connected to 203.230.17.254.  
sftp> ls  
Desktop          Documents        Downloads        Music            Pictures         Public          Templates  
Videos           ca.crt           dh2048.pem      snap            ta.key          test.conf      test.crt  
test.key         ubuntu01.crt    ubuntu01.key    ubuntu01.ovpn  
sftp> mget ca.*  
Fetching /home/ubuntu/ca.crt to ca.crt  
/home/ubuntu/ca.crt          100% 1204      2.0MB/s   00:00  
sftp> mget ta.*  
Fetching /home/ubuntu/ta.key to ta.key  
/home/ubuntu/ta.key          100% 636       1.2MB/s   00:00  
sftp> mget ubuntu01.*  
Fetching /home/ubuntu/ubuntu01.crt to ubuntu01.crt  
/home/ubuntu/ubuntu01.crt    100% 4498     9.3MB/s   00:00  
Fetching /home/ubuntu/ubuntu01.key to ubuntu01.key  
/home/ubuntu/ubuntu01.key    100% 1704     2.0MB/s   00:00  
Fetching /home/ubuntu/ubuntu01.ovpn to ubuntu01.ovpn  
/home/ubuntu/ubuntu01.ovpn   100% 3593     5.6MB/s   00:00  
sftp> exit  
ubuntu@client2:~$ ls  
ca.crt  Documents  Music  Public  Templates  ubuntu01.key  Videos  
Desktop Downloads Pictures ta.key  ubuntu01.crt  ubuntu01.ovpn  
ubuntu@client2:~$
```

OpenVPN 통신

■ 클라이언트 실행

➤ 현재 IP 확인

```
ubuntu@client2: ~  
ubuntu@client2:~$ ifconfig  
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 203.230.17.10  netmask 255.255.255.0  broadcast 203.230.17.255  
    inet6 fe80::a00:27ff:fe22:4811  prefixlen 64  scopeid 0x20<link>  
    ether 08:00:27:22:48:11  txqueuelen 1000  (Ethernet)  
    RX packets 2401  bytes 2917802 (2.9 MB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 2358  bytes 202425 (202.4 KB)  
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536  
    inet 127.0.0.1  netmask 255.0.0.0  
    inet6 ::1  prefixlen 128  scopeid 0x10<host>  
    loop txqueuelen 1000  (Local Loopback)  
    RX packets 1660  bytes 141069 (141.0 KB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 1660  bytes 141069 (141.0 KB)  
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0  
  
ubuntu@client2:~$
```

OpenVPN 통신

■ 클라이언트 실행

➤ OpenVPN 클라이언트 실행

```
ubuntu@client2: ~  
ubuntu@client2:~$ sudo openvpn --config ubuntu01.ovpn  
[sudo] password for ubuntu:  
Mon Dec 12 00:00:36 2022 WARNING: file 'ubuntu01.key' is group or others accessible  
Mon Dec 12 00:00:36 2022 WARNING: file 'ta.key' is group or others accessible  
Mon Dec 12 00:00:36 2022 OpenVPN 2.4.7 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] built on Mar 22 2022  
Mon Dec 12 00:00:36 2022 library versions: OpenSSL 1.1.1f 31 Mar 2020, LZO 2.10  
Mon Dec 12 00:00:36 2022 Outgoing Control Channel Authentication: Using 160 bit message hash 'SHA1' for HMAC authentication  
Mon Dec 12 00:00:36 2022 Incoming Control Channel Authentication: Using 160 bit message hash 'SHA1' for HMAC authentication  
Mon Dec 12 00:00:36 2022 TCP/UDP: Preserving recently used remote address: [AF_INET]203.230.17.254:1194  
Mon Dec 12 00:00:36 2022 Socket Buffers: R=[212992->212992] S=[212992->212992]  
Mon Dec 12 00:00:36 2022 UDP link local: (not bound)  
Mon Dec 12 00:00:36 2022 UDP link remote: [AF_INET]203.230.17.254:1194  
Mon Dec 12 00:00:36 2022 TLS: Initial packet from [AF_INET]203.230.17.254:1194, sid=dd9550b3 31649c20  
Mon Dec 12 00:00:36 2022 VERIFY OK: depth=1, CN=Easy-RSA CA  
Mon Dec 12 00:00:36 2022 VERIFY KU OK  
Mon Dec 12 00:00:36 2022 Validating certificate extended key usage  
Mon Dec 12 00:00:36 2022 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authentication  
Mon Dec 12 00:00:36 2022 VERIFY EKU OK  
Mon Dec 12 00:00:36 2022 VERIFY OK: depth=0, CN=test  
Mon Dec 12 00:00:36 2022 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, 2048 bit RSA  
Mon Dec 12 00:00:36 2022 [test] Peer Connection Initiated with [AF_INET]203.230.17.254:1194  
Mon Dec 12 00:00:37 2022 SENT CONTROL [test]: 'PUSH_REQUEST' (status=1)
```

OpenVPN 통신

■ 클라이언트 실행

- OpenVPN 클라이언트 실행 후 IP 확인
- Tun0 인터페이스 및 IP주소 확인

```
ubuntu@client2: ~  
ubuntu@client2:~$ ifconfig  
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 203.230.17.10 netmask 255.255.255.0 broadcast 203.230.17.255  
    inet6 fe80::a00:27ff:fe22:4811 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:22:48:11 txqueuelen 1000 (Ethernet)  
    RX packets 2451 bytes 2927226 (2.9 MB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2417 bytes 211543 (211.5 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 1710 bytes 147179 (147.1 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1710 bytes 147179 (147.1 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
tun0: flags=4305<UP,POINTOPOINT,RUNNING,NOARP,MULTICAST> mtu 1500  
    inet 10.8.0.6 netmask 255.255.255.255 destination 10.8.0.5  
    inet6 fe80::1c7e:5d79:ad0b:12f3 prefixlen 64 scopeid 0x20<link>  
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 100 (UNSPEC)  
    RX packets 14 bytes 1013 (1.0 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 19 bytes 1113 (1.1 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

OpenVPN 통신

■ 클라이언트 실행

- 라우팅 테이블 확인
- 내부 네트워크인 192.168.1.0/24 의 라우팅이 tun0으로 변경 확인

```
ubuntu@client2: ~  
ubuntu@client2:~$ netstat -rn  
Kernel IP routing table  
Destination      Gateway          Genmask          Flags      MSS  Window  irtt  Iface  
0.0.0.0          203.230.17.1    0.0.0.0          UG          0 0      0     enp0s3  
10.8.0.1         10.8.0.5        255.255.255.255 UGH          0 0      0     tun0  
10.8.0.5         0.0.0.0         255.255.255.255 UH          0 0      0     tun0  
169.254.0.0      0.0.0.0         255.255.0.0      U           0 0      0     enp0s3  
192.168.1.0      10.8.0.5        255.255.255.0    UG          0 0      0     tun0  
203.230.17.0     0.0.0.0         255.255.255.0    U           0 0      0     enp0s3  
ubuntu@client2:~$
```


OpenVPN 통신

■ 통신 테스트

➤ Client2 → Server1 로 통신 요청

```
ubuntu@client2: ~  
ubuntu@client2:~$ ping 192.168.1.10 -c 2  
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.  
64 bytes from 192.168.1.10: icmp_seq=1 ttl=63 time=1.02 ms  
64 bytes from 192.168.1.10: icmp_seq=2 ttl=63 time=0.777 ms  
  
--- 192.168.1.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1099ms  
rtt min/avg/max/mdev = 0.777/0.897/1.017/0.120 ms  
ubuntu@client2:~$
```

➤ Server1에서의 TCPDUMP 화면

```
root@Server1: ~  
root@Server1:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
00:05:31.620347 IP 10.8.0.6 > Server1: ICMP echo request, id 3, seq 1, length 64  
00:05:31.620370 IP Server1 > 10.8.0.6: ICMP echo reply, id 3, seq 1, length 64
```

➤ 클라이언트의 tun0 IP인 10.8.0.6으로 통신 확인

OpenVPN 통신

■ 통신 테스트

➤ OpenVPN 서버에서의 TCPDUMP 확인

```
root@OpenVPN: ~  
root@OpenVPN:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on tun0, link-type RAW (Raw IP), capture size 262144 bytes  
00:08:21.783417 IP 10.8.0.6 > 192.168.1.10: ICMP echo request, id 4, seq 1, length 64  
00:08:21.783918 IP 192.168.1.10 > 10.8.0.6: ICMP echo reply, id 4, seq 1, length 64  
00:08:22.815905 IP 10.8.0.6 > 192.168.1.10: ICMP echo request, id 4, seq 2, length 64  
00:08:22.816119 IP 192.168.1.10 > 10.8.0.6: ICMP echo reply, id 4, seq 2, length 64  
^C  
4 packets captured  
4 packets received by filter  
0 packets dropped by kernel  
root@OpenVPN:~#
```

End of Document